

Koushik Bhupathiraju

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Education

Masters of Science in Computer Science, Quinnipiac University Jan 2025 – Dec 2026
Algorithms and Design, Operating Systems, Database Systems, Embedded Systems | **GPA:** 3.77/4.0 Hamden, CT

B.Tech in Electronics and Communication, National Institute of Technology Jul 2018 – May 2022
Data Structures, Computer Programming, Neural Networks and Deep Learning Karnataka, India

Technical Skills

Languages: Python, SQL, HTML5, CSS, JavaScript, C, C++, Java

Technologies: Hadoop, Hive, Spark, Cassandra, NoSQL, Object-oriented programming, Informatica PowerCenter

Developer Tools: AWS DynamoDB, Postman, GIT, MongoDB, Oracle SQL Developer, Jupyter, VS Code

Libraries/Frameworks: ReactNative, ReactJS, MySQL, NodeJS, PySpark, Pandas, MediaPipe, Threading

AI/ML: LLMs, OpenCode, ClaudeCode, AI Agents, Inference SDK, Transformers, Ollama, VectorDB, MCP

Work Experience

Software Engineer Oct 2022 – Dec 2024
Accenture Solutions, Inc Hyderabad, India

- Developed an internal **chatbot** to assist teams with workflow status checks, job failure insights, and common operational queries, improving response time and reducing manual effort. Built using **Python**, REST APIs.
- Monitored and resolved transmission issues from SPQ to **Informatica PowerCenter** and **Oracle SQL Developer**, ensuring seamless data flow to ERPs like **SAP**, **Novora**, and **Ariba**.
- Queried and debugged job failures in **Oracle SQL Developer**, improving transmission reliability across daily and weekly batch jobs. Optimized and rewrote repetitive **SQL** queries, improving maintainability and performance.
- Handled duplicate and missing data issues by analyzing workflows and ERPs, improving data accuracy and consistency by 30%. Used tools like **SOAP UI** and **Postman** for backend data testing and validation.
- Built ingestion pipelines with **Kafka** and **HDFS**, storing data in **Cassandra**, **MongoDB**, and **Hadoop**.

Software Development Intern May 2025 – Aug 2025
Rammina Dev & Design Remote, Philippines

- Built a fundraiser web app using **React.js**, **Node.js/Express**, and **MongoDB**.
- Integrated **Stripe** for secure payment processing and used **Google OAuth** for seamless user authentication.
- Utilized **JWT** tokenization for secure user sessions and built RESTful APIs, tested with **Postman**.
- Leveraged **Cloudinary** for efficient media storage and management, ensuring optimized image handling.

Graduate Assistant May 2025 – Present
Quinnipiac University Hamden, Connecticut

- Advised international applicants on admissions, visa processes, and program selection across multiple regions.
- Automated updates to the visa issuance Excel sheet using **Pandas** and generated insights through charts.
- Reviewed academic transcripts and grading scales to assist accurate and timely international application decisions.

Projects

Thesis | *Energy Benchmarking of KV Cache Compression Methods on GPU*

- Completed a thesis on energy benchmarking of **KV cache compression** for LLM inference on GPUs.
- Built a GPU power measurement pipeline using NVML to compute joules/token and tokens/joule.
- Benchmarked **eviction**, **quantization**, and hybrid KV cache methods against no-compression baselines.
- Evaluated energy-performance trade-offs across context lengths, batch sizes, and models.
- Analyzed the impact of compression ratio vs. compute overhead on energy efficiency.
- Produced reproducible benchmarks and practical guidelines for energy-efficient **LLM deployment**.

Money-Management App | *ReactNative, Nodejs, ExpressJs*

- Developed a cross-platform money management app using **Expo** for rapid prototyping and deployment.
- Built a native-like front end with **ReactNative**, ensuring a seamless user experience across platforms.
- Implemented efficient server-side logic with **Node.js**, handling asynchronous requests effectively.
- Deployed and managed backend services on **AWS DynamoDB**, ensuring security and scalability.